AUSTIN DECICCO

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EDUCATION

Rutgers University, New Brunswick, NJ
M.S. in Mathematics — GPA 4.0/4.0 *Expected May 2027*

Rutgers University, New Brunswick, NJ
B.A. in Mathematics — GPA 3.96/4.0, ***Summa Cum Laude*** *May 2023*

University of Maryland, College Park, MD
B.S. in Economics — Minor in Statistics *May 2022*

RESEARCH INTERESTS

Quantitative finance; Stochastic investment models; Regime-switching models; Reinforcement Learning; Econometrics; Combinatorics.

RESEARCH

Liquidity-Dependent Retrace of Price Spikes under Cross-Venue Arbitrage
Independent research, supervised by Prof. Larsen, Rutgers *Fall 2026*

- Proposed a model of how far a CEX price spike retraces once arbitrage with a constant-product AMM pool closes the gap, as a function of pool liquidity, slippage, and book depth.

Statistical Properties of Permutations and SYT Reflected by the Robinson–Schensted Algorithm
With Jike Liu — Experimental Mathematics (Prof. Zeilberger), Rutgers *Spring 2026*

- Ran row/column-sum-vector uniqueness experiments on standard Young tableaux, finding two integer sequences not previously in the OEIS (computed through $n = 13$).
- Computational study of the RS bijection in Maple: verified the Logan–Shepp limit shape and the $\lambda_1/\sqrt{n} \to 2$ convergence for the longest increasing subsequence.
- Quantified the covariance and correlation of longest increasing / decreasing subsequence lengths, and tested an asymptotic-independence conjecture for fixed-entry locations in the insertion and recording tableaux.
- Generated bootstrap datasets ($n = 1$ to 30, up to $3{,}000$ samples per n) and consolidated the Maple source; paper draft, slides, code, and data at austindecicco.com/gradp.html.

SELECTED COURSEWORK

Graduate (in progress, Fall 2026): Theory of Functions of a Real Variable, Abstract Algebra, Combinatorics, Algorithmic Trading and Portfolio Management, Financial Statistics and Risk Management.

Graduate (Rutgers, completed): Experimental Mathematics (A), Graph Theory (A), Introduction to Mathematical Physics (A), Probability in Discrete Mathematics (A).

Undergraduate (Rutgers / Maryland): Introduction to Stochastic Processes (A), Mathematical Theory of Statistics (A), Advanced Topics in Econometrics (A), Econometrics (A+), Game Theory (A).

Experience

Independent Generalist Investor

U.S. Equity Markets; Residential Property *May 2023 – Present*

- Special-situations and thematic equity. Largest successes were a post-restructuring position in American Coastal Insurance (ACIC, formerly UIHC) and thematic AI and semiconductor exposure during the 2023–2024 rally.
- Directed the renovation of a residential property: managed contractors, capital improvements, and project accounting.

Independent Cryptocurrency Arbitrage

Ethereum and VeChain DeFi Ecosystems *2020 – 2022*

- Manually executed hundreds of cross-DEX arbitrage trades on Ethereum (Uniswap) and VeChain (Vexchange, Betterswap, and others) during the 2020–2022 DeFi cycle; the entry threshold scaled with capital and gas to clear fixed and proportional fees plus a buffer for expected spread decay over the $\sim$5-minute manual execution window.

Technical Skills

Programming: Python, Maple, Java, SAS, Stata, MATLAB, LaTeX, HTML, CSS.

Quantitative: stochastic processes, mathematical statistics, time-series and panel-data econometrics, regression and hypothesis testing, combinatorics, Monte Carlo and bootstrap methods.

Markets: 10+ years personal trading (equities, options, crypto); fundamental and special-situations equity analysis.

Service & Outreach

Volunteer Firefighter — Flemington Fire Department, NJ *May 2024 – Present*

Executive, College Park Student Investment Group

University of Maryland *2020 – 2021*

- Led group meetings, prepared market commentary, and taught peers equity-market and trading mechanics.